

Lecture 11

VLSI Interconnect

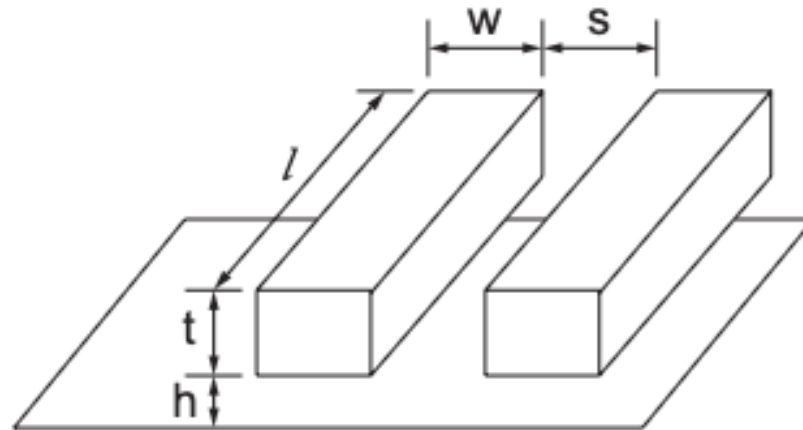
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Interconnect Wires

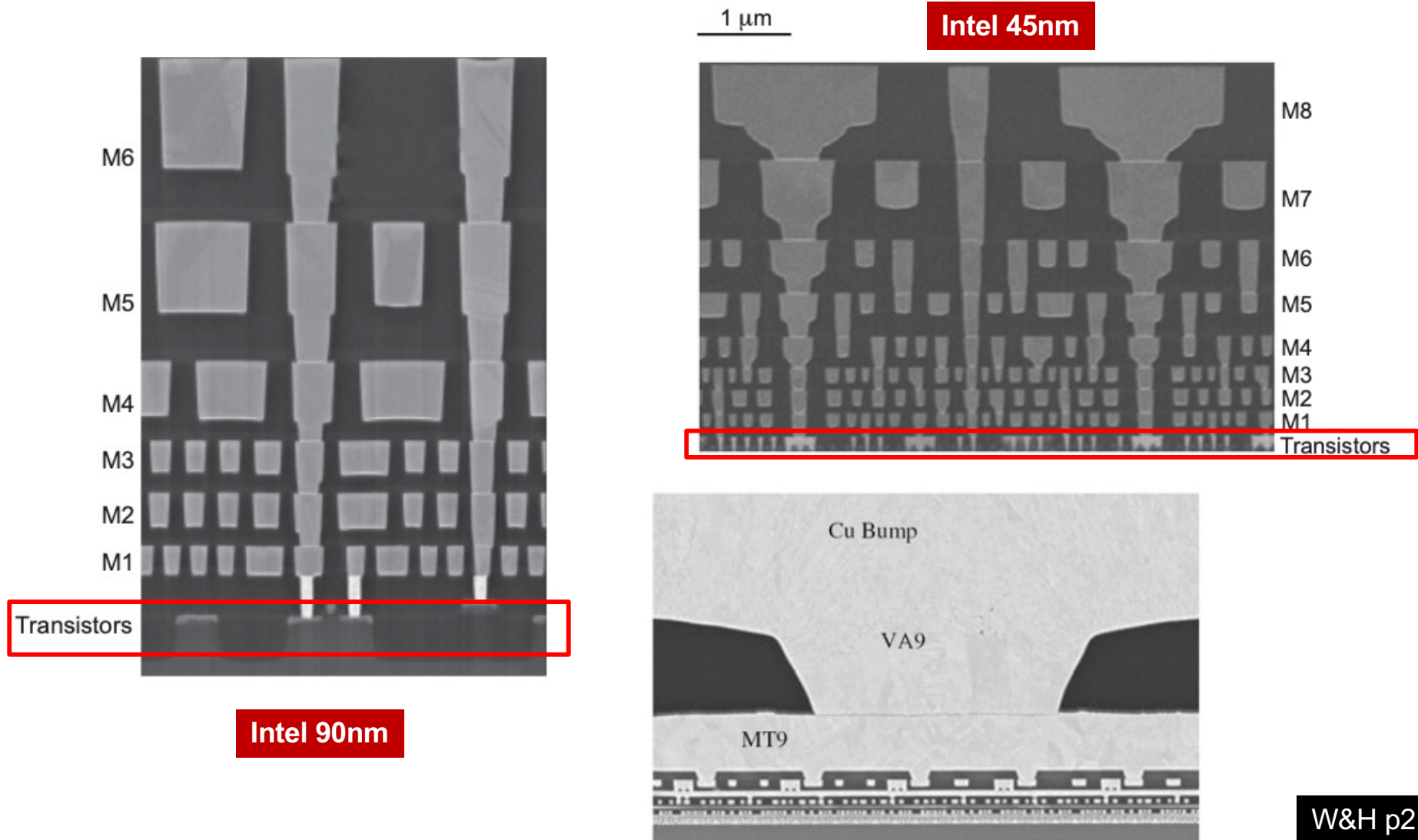
- ◆ **Wires** dominates VLSI chips.
- ◆ As feature sizes shrink → wires as important as transistors.
- ◆ Wires affect: 1) Speed; 2) Power; 3) Noise.
- ◆ Always use alternating layers at right angle to each other.
- ◆ Wire's pitch = $w + s$.
- ◆ Wire's aspect ratio $AR = t/w$.
- ◆ Modern process $AR \approx 2$.



Metal Layer Stacks

- ◆ A typical 0.6 μ m process has 3 metal layers:
 - ❖ M1 for within cell routing
 - ❖ M2 for vertical routing between cells
 - ❖ M3 for horizontal routine between cells.
- ◆ Modern processes use 6 to 10+ metal layers:
 - ❖ M1 for high density cells – thin and narrow
 - ❖ Middle layers (M2 up) – thicker and wider for routing between cells
 - ❖ Top few layers - for VDD, GND and clock signals.

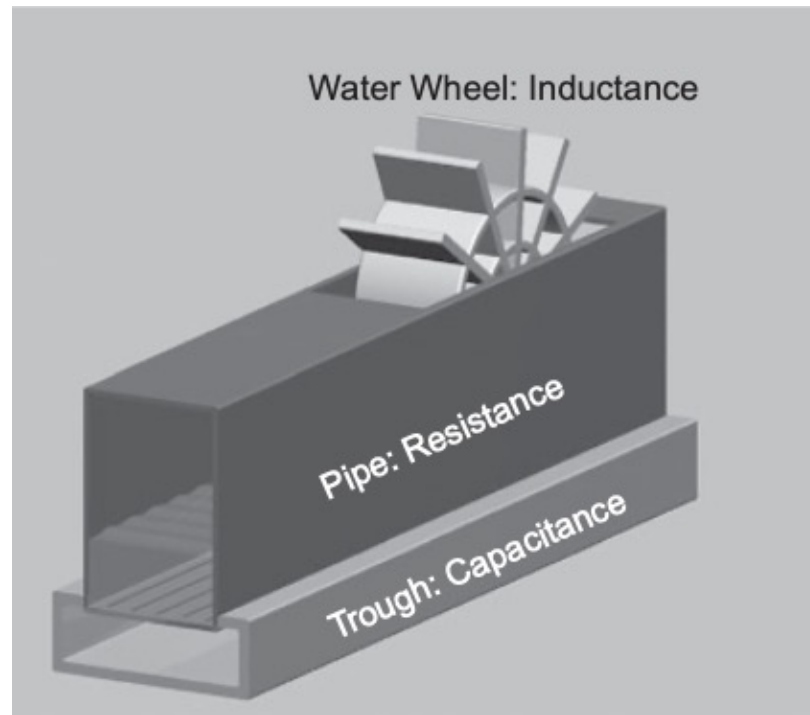
SEM images of metal stack



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Interconnect Modelling – waterpipe analogy

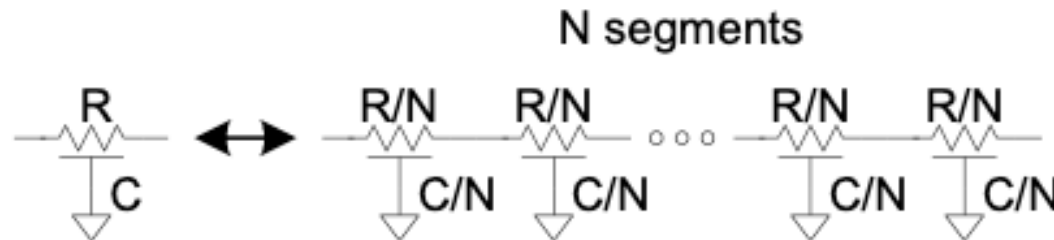
- ◆ Current in a wire is analogous to water in a pipe.
 - ❖ **Resistance**: narrow size impedes the flow of water.
 - ❖ **Capacitance**: trough under the leaky pipe must fill first.
 - ❖ **Inductance**: paddle wheel inertia opposes changes in flow rate (negligible except for long wires).



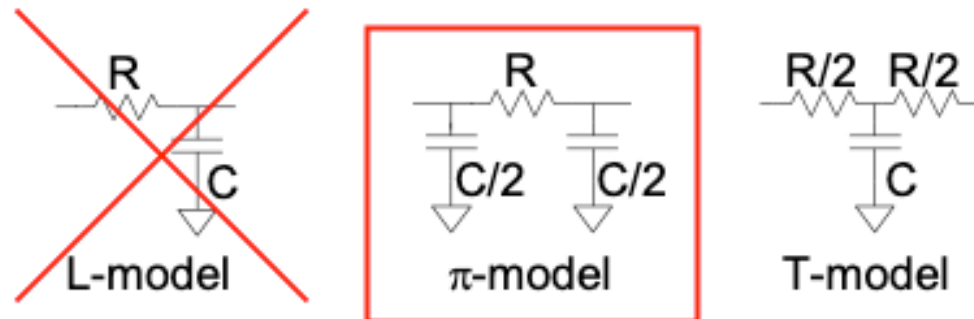
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Interconnect Modelling – Lumped model

- Wires are distributed circuits approximated by a **lumped element model**.



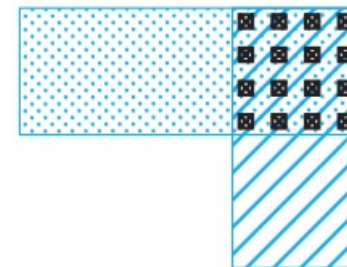
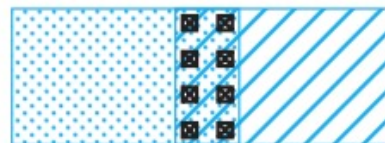
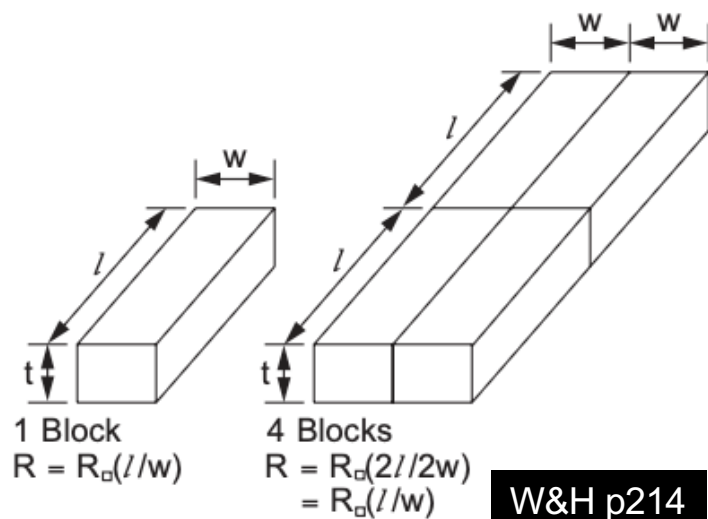
- 3-segment of π -model is accurate to 3% in simulation.
- L-model requires 100+ segments to achieve same accuracy.
- Use single 3-segment π -model for Elmore delay calculation.



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Interconnect Resistance

- ◆ ρ = resistivity ($\Omega \cdot \text{m}$).
- ◆ Resistance $R = \frac{\rho}{t} \times \frac{l}{w}$.
- ◆ Sheet Resistance $R_{sq} = \frac{\rho}{t}$, dimension is ohms per square.
- ◆ Count number of squares: $R = R_{sq} \times \text{no. of squares}$
- ◆ Two wires below have same resistance.
- ◆ Polysilicon and diffusion sheet resistance: much higher at $10 \Omega/\text{sq}$ to $100\text{s } \Omega/\text{sq}$.
- ◆ Contact and vias typically has $2 - 20 \Omega$.

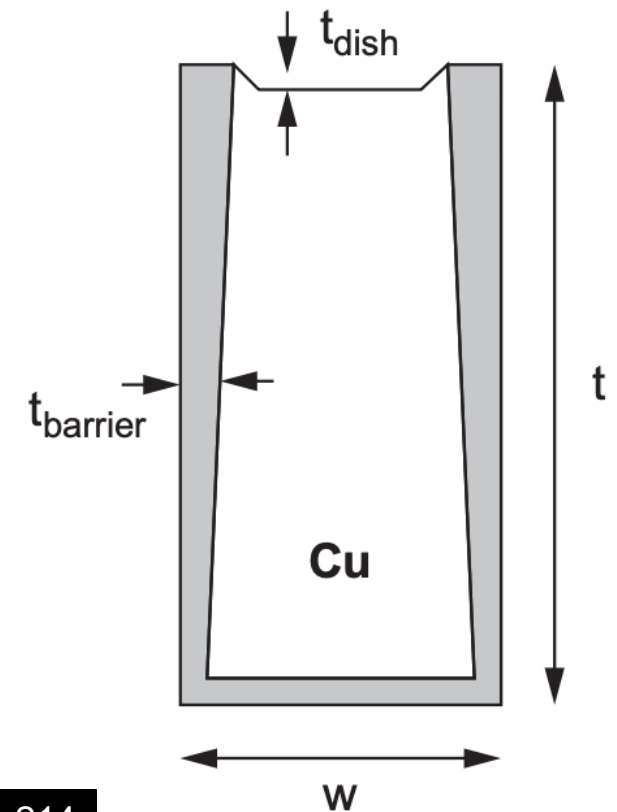


Copper wires

- ◆ Copper wires diffusion barrier has high resistance
- ◆ Copper is also prone to *dishing* during polishing
- ◆ Effective resistance is higher

$$R = \frac{\rho}{\left(t - t_{\text{dish}} - t_{\text{barrier}}\right)} \frac{l}{\left(w - 2t_{\text{barrier}}\right)}$$

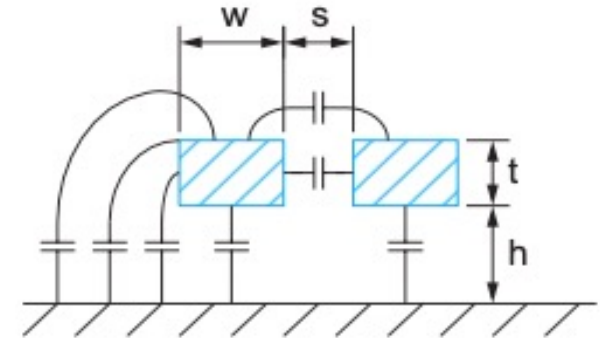
Metal	Resistivity ($\mu\Omega \cdot \text{cm}$)
Silver (Ag)	1.6
Copper (Cu)	1.7
Gold (Au)	2.2
Aluminum (Al)	2.8
Tungsten (W)	5.3
Molybdenum (Mo)	5.3
Titanium (Ti)	43.0



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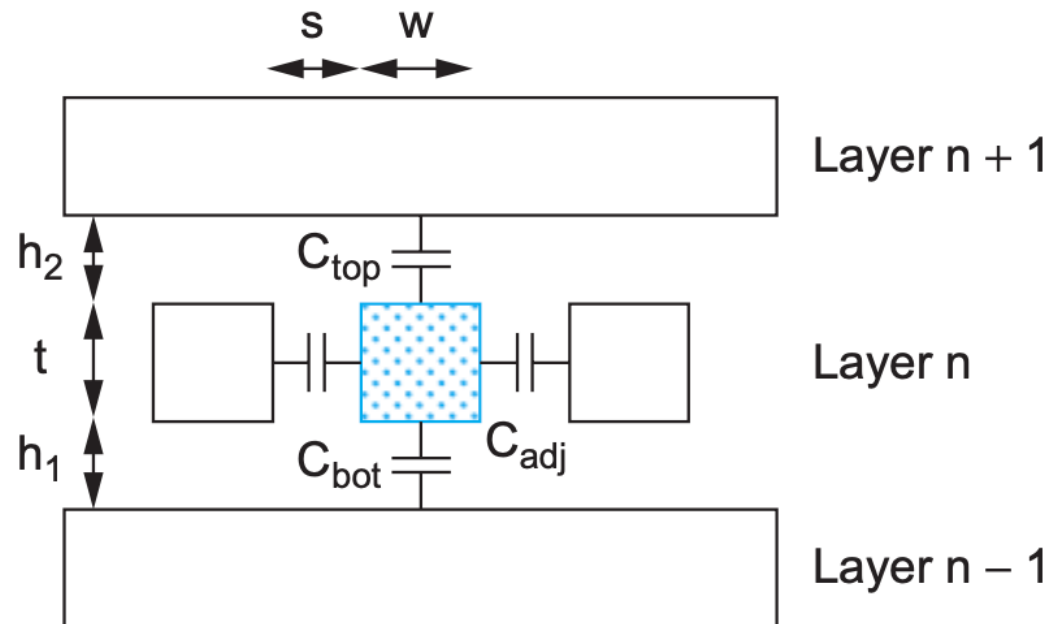
Wire capacitances

- ◆ Parallel plate equation:
$$C = \frac{\epsilon_{ox}}{h} wl$$
- ◆ Wires are not parallel plates
 - Increasing area (W, t) increases capacitance
 - Increasing distance (s, h) decreases capacitance
- ◆ Dielectric constant
 - $\epsilon_{ox} = k\epsilon_0$
 - $\epsilon_0 = 8.85 \times 10^{-14}$ F/cm
 - $k = 3.9$ for SiO_2
- ◆ Processes are starting to use low-k dielectrics
 - $k \approx 3$ (or less)



Wire capacitances

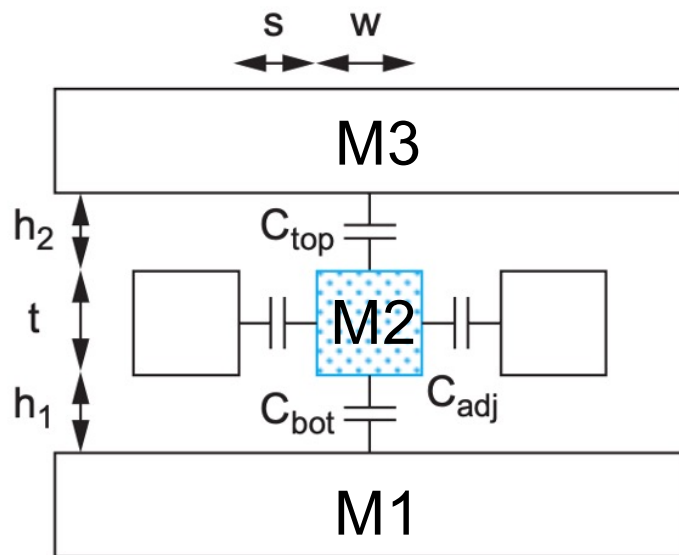
- ◆ Wire has capacitance per unit length
 - To neighbours
 - To layers above and below
- ◆ $C_{\text{total}} = C_{\text{top}} + C_{\text{bot}} + 2C_{\text{adj}}$



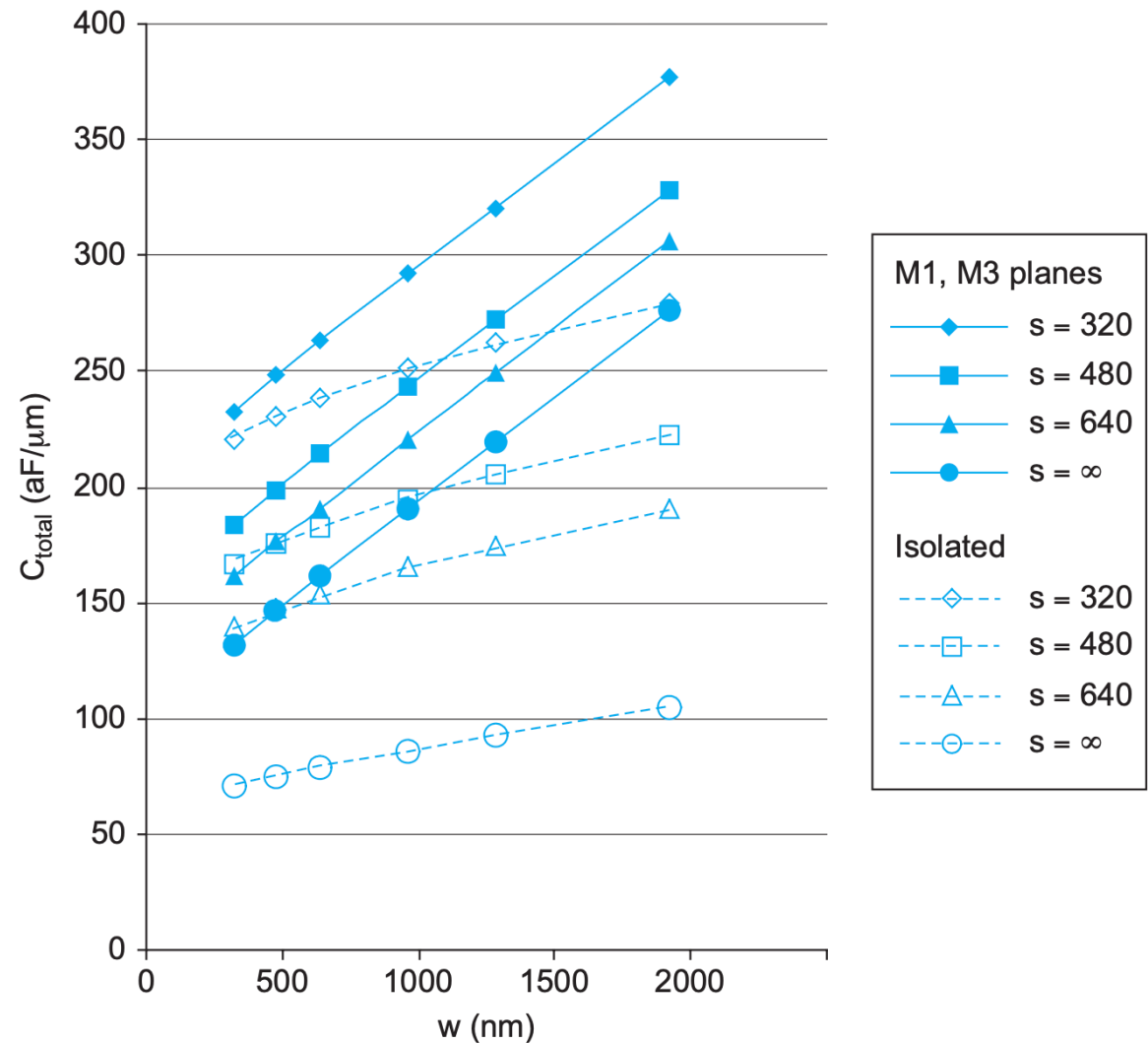
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Metal 2 capacitance

- ◆ Typical dense wires have $\sim 0.2 \text{ fF}/\mu\text{m}$
- ◆ Compare to 1-2 $\text{fF}/\mu\text{m}$ for gate capacitance



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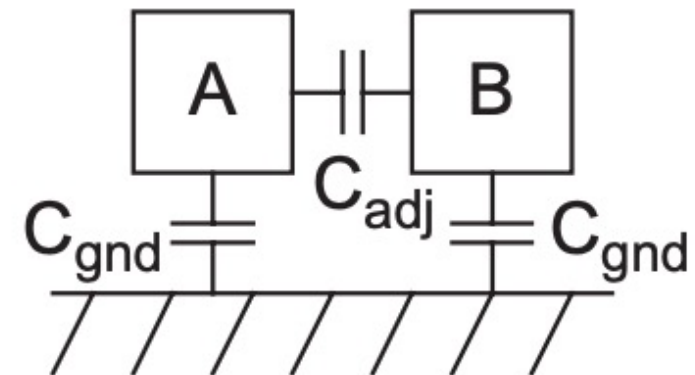


Diffusion and Polysilicon Wires

- ◆ Diffusion capacitance is very high ($1\text{-}2\text{ fF}/\mu\text{m}$)
 - Comparable to gate capacitance
 - Diffusion also has high resistance
 - Avoid using diffusion *runners* for wires!
- ◆ Polysilicon has lower C but high R
 - Use for transistor gates
 - Occasionally for very short wires between gates

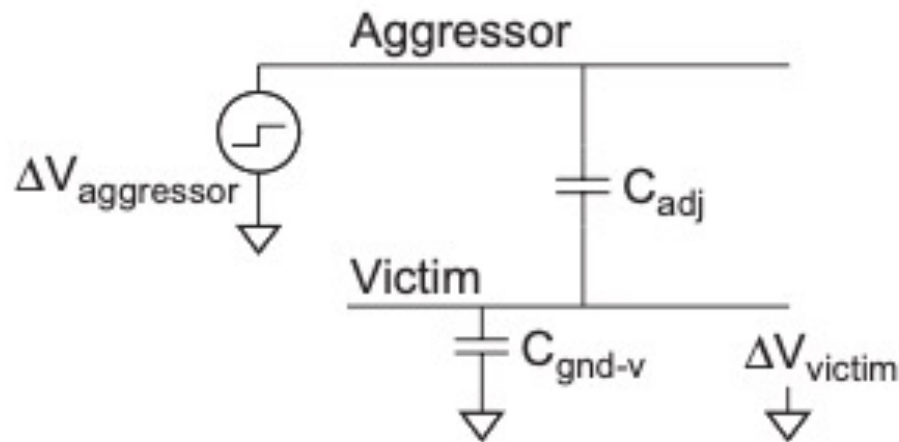
Wire Crosstalk

- ◆ A capacitor does not like to change its voltage instantaneously.
- ◆ A wire has high capacitance to its neighbour.
 - When the neighbour switches from 1 $\rightarrow$ 0 or 0 $\rightarrow$ 1, the wire tends to switch too.
 - Called capacitive *coupling* or *crosstalk*.
- ◆ Crosstalk causes
 - Noise on non-switching wires
 - Increased delay on switching wires



Crosstalk Noise – floating victim

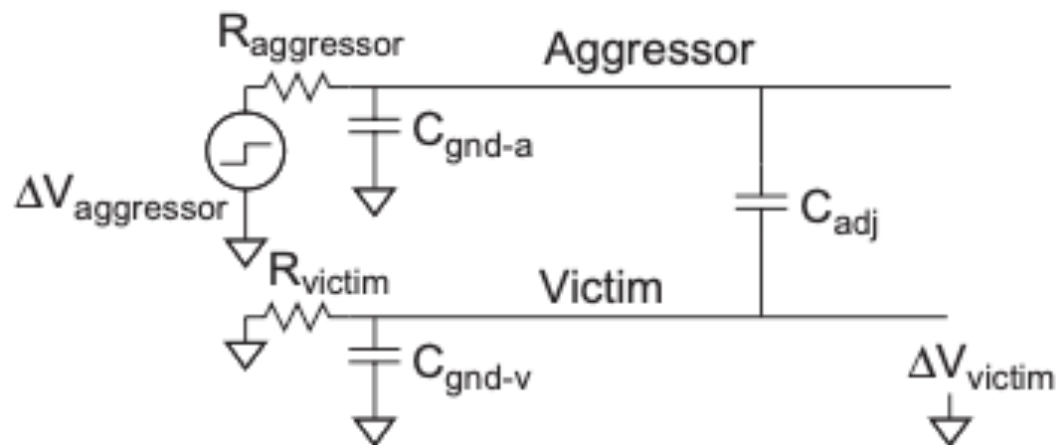
- ◆ Crosstalk causes noise on non-switching wires
- ◆ If victim is floating:
 - model as capacitive voltage divider



$$\Delta V_{\text{victim}} = \frac{C_{\text{adj}}}{C_{\text{gnd-v}} + C_{\text{adj}}} \Delta V_{\text{aggressor}}$$

Crosstalk Noise – driven victim

- ◆ Usually victim is driven by a gate that fights noise
 - Noise depends on relative resistances
 - Having driven victim increase k and reduces crosstalk noise
 - Avoid floating wires to minimize crosstalk



$$\Delta V_{victim} = \frac{C_{adj}}{C_{gnd-v} + C_{adj}} \frac{1}{1+k} \Delta V_{aggressor}$$

$$k = \frac{\tau_{aggressor}}{\tau_{victim}} = \frac{R_{aggressor} (C_{gnd-a} + C_{adj})}{R_{victim} (C_{gnd-v} + C_{adj})}$$

Noise Coupling Waveforms

- ◆ Simulated coupling for $C_{adj} = C_{victim}$



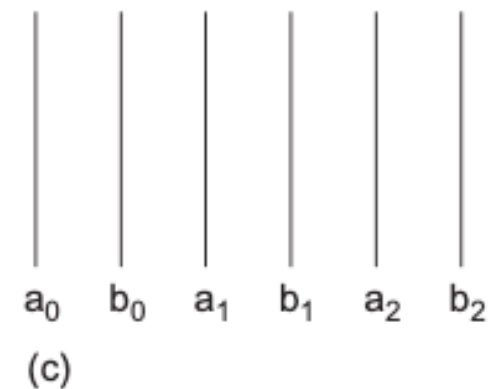
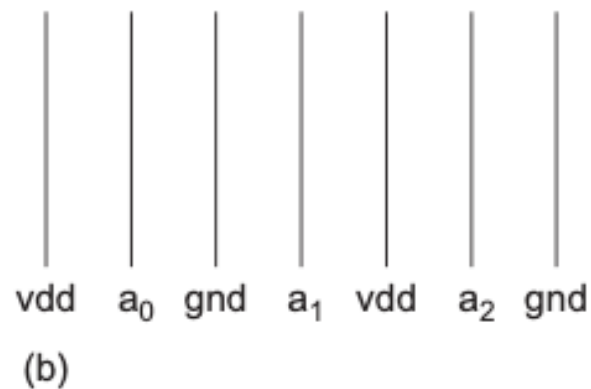
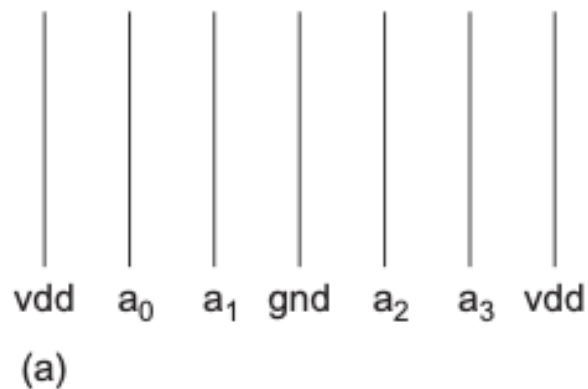
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So, what if we have noise?

- ◆ If the noise is less than the noise margin, nothing happens.
- ◆ Static CMOS logic will eventually settle to correct output even if disturbed by large noise spikes.
 - Glitches cause extra delay.
 - Glitches cause extra power from false transitions.
- ◆ Dynamic logic never recovers from glitches.
- ◆ Memories and other sensitive circuits also can produce the wrong answer

Crosstalk Control via Shielding

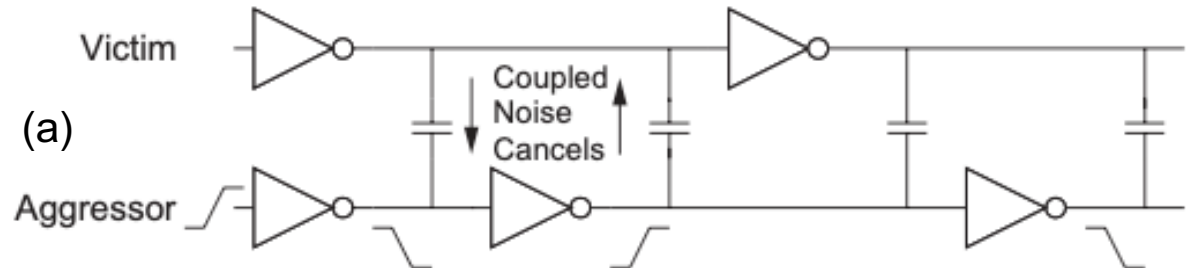
- ◆ Increase spacing to adjacent wires
- ◆ Shield wires
- ◆ Ensure neighbours switch at different times
- ◆ Crosstalk cancellation



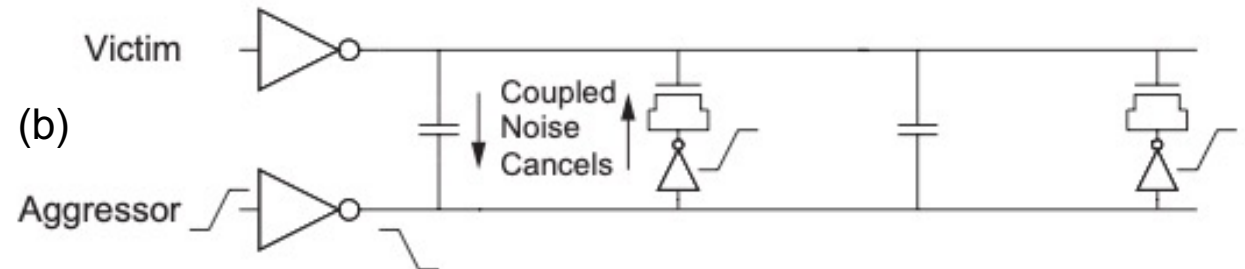
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Crosstalk Control via Cancellation

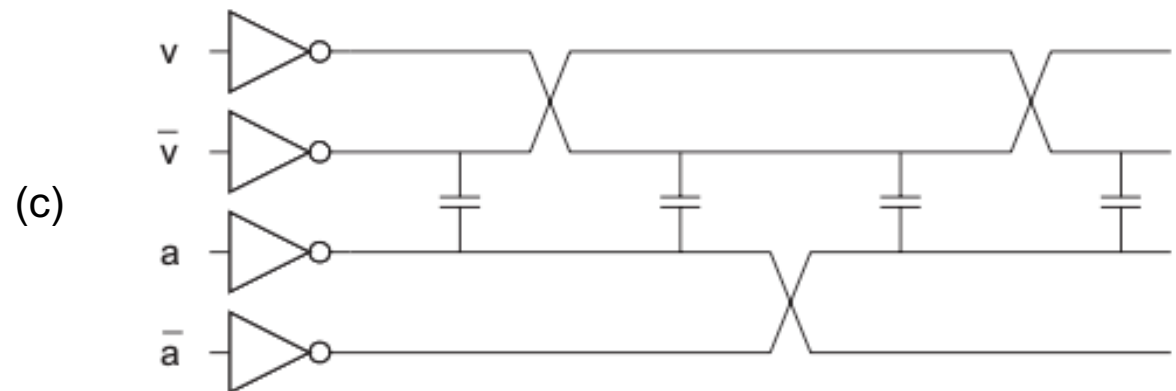
- ◆ Staggered repeaters:



- ◆ Charge compensation:



- ◆ Twisted differential signalling:



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